Sayantan Pal

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Education

Jalpaiguri Government Engineering College **11/2021-05/2025**
Jalpaiguri, West Bengal, India, Pin:- 735102
Bachelor of Technology in Electronics and Communication Engineering **YGPA: 7.647/10.0**
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Object-Oriented Design Patterns, Database Systems

Experience

Associate Software Engineer Hybrid (02/2025-Present)
Prismforce, *Pune, Maharashtra, India*
- Collaborated with cross-functional stakeholders to architect **SelectPrism** from scratch—a recruitment platform built on **Node.js, TypeScript, Python (FastAPI), MongoDB, and AWS**, now reliably handling 1,000+ concurrent interview sessions in production.
- Reduced voice AI interview latency to enable real-time conversations by integrating **WebRTC and LiveKit**, then optimized the STT→LLM→TTS pipeline through async batching and profiling to achieve **P95 latency under 200ms**.
- Developed a **resume parser using Ollama with local LLMs**, applying **data structure optimizations** for field extraction; validated against 500 manually-labeled resumes, achieving **92% accuracy** across 12 critical fields (name, skills, experience, education).
- Addressed authentication vulnerabilities identified during security audit by implementing **Redis-based rate limiting** (100 req/min per IP), bcrypt hashing (cost factor 12), and Google reCAPTCHA—passed re-assessment with zero critical findings.
- Designed **extensible, maintainable automation systems**: IVR campaigns via Ozonetel and a **Bull queue-based email service**, scaling candidate outreach from 100 contacts/3 days to **5,000+ daily notifications**.
- Improved API **reliability and observability** by adding MongoDB schema validation and structured error logging; implemented **RTK Query caching** on frontend, reducing average response time from 340ms to 220ms.

Data Science Intern Remote (05/2024-07/2024)
Celebal Technologies, *Jalpaiguri, WB, India*
- Analyzed attrition patterns across salary bands and tenure cohorts; discovered mid-tenure employees (3–5 years) earning 27L–40L showed 40% lower turnover using **K-means clustering** and statistical analysis.
- Presented salary-vs-experience retention analysis to HR leadership, proposing data-driven pay-band restructuring recommendations.

Projects

AI-Powered Resume Matching Engine — **Python, FastAPI, Ollama, MongoDB, Docker** *01/2025-02/2025*
- Designed and implemented an end-to-end pipeline: PDF ingestion → text extraction → embedding generation via Ollama → **vector similarity search** in MongoDB → ranked candidate API responses.
- Collaborated with recruiters to create ground-truth labels for 200 job-resume pairs; system's top-5 recommendations matched the correct candidate **88% of the time**.
- Built with **diagnosability in mind**: integrated structured logging, health checks, and **LocalStack** for local AWS simulation (S3, SQS), reducing cloud costs during development.

Tennis Analysis System with YOLO and Keypoint Extraction *01/2025-01/2025*
- Fine-tuned **YOLOv8** on a custom tennis dataset—maintains consistent player tracking across 1,200+ frames without ID loss during rallies.
- Trained a **PyTorch CNN** using efficient **data structures** for batched inference, detecting 17 court keypoints per frame with 92% accuracy; used **OpenCV** for preprocessing and visualization.
- Integrated detection, tracking, and keypoint models into a unified, **maintainable pipeline** that processes raw match footage to output player positions and court geometry.

Technical Skills

- **Languages:** Python, JavaScript, TypeScript, C, C++, Verilog, HTML, CSS
- **Backend:** Node.js, Express.js, FastAPI, Flask, REST APIs, WebRTC, LiveKit
- **Frontend:** React.js, Next.js, Redux Toolkit, RTK Query, Bootstrap
- **Databases:** MongoDB (indexing, aggregation), Redis, SQL
- **ML/AI:** PyTorch, TensorFlow, Scikit-learn, Ollama, YOLOv8, Pandas, NumPy
- **DevOps/Cloud:** AWS (EC2, S3, SQS, Lambda), Docker, Apache Airflow, Git, CI/CD
- **Core CS:** Data Structures, Algorithms, Design Patterns (Factory, Observer, MVC), System Design

Achievements

- Presented **An Advanced Framework For Cardiac Risk Prediction And Real-Time Monitoring Using Machine Learning And IoT** at **ICDEC-2025**—designed a system that flags cardiac risk in real-time using ML on IoT sensor data.
- Published work on an **FPGA-based timing generator** at **NCRTST-2025**—achieved timing precision of ±1ns for cold collision experiments.
- Won $100 at **Quine Quest 22** for an AI-powered document summarizer.